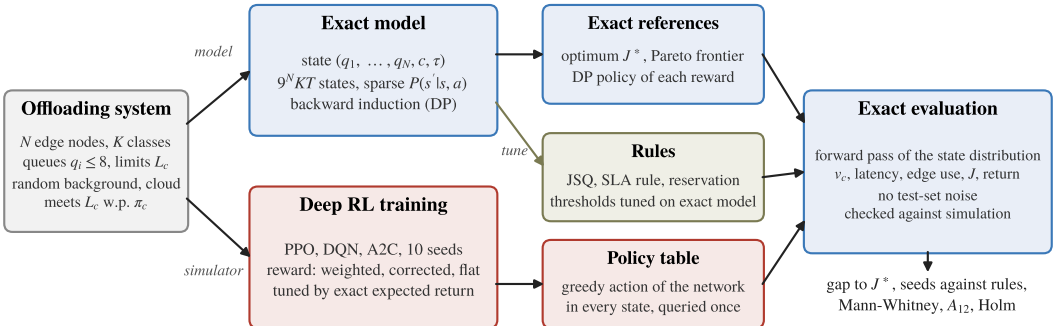
Offloading system
N edge nodes, K classes
queues $q_i \leq 8$, limits L_c
random background, cloud
meets L_c w.p. π_c
model
simulator
Exact model
state $(q_1, \ldots, q_N, c, \tau)$
$9^N KT$ states, sparse $P(s'|s, a)$
backward induction (DP)
Deep RL training
PPO, DQN, A2C, 10 seeds
reward: weighted, corrected, flat
tuned by exact expected return
Exact references
optimum J^*, Pareto frontier
DP policy of each reward
tune
Rules
JSQ, SLA rule, reservation
thresholds tuned on exact model
Policy table
greedy action of the network
in every state, queried once
Exact evaluation
forward pass of the state distribution
v_c, latency, edge use, J, return
no test-set noise
checked against simulation
gap to J^*, seeds against rules,
Mann-Whitney, A_{12}, Holm